

Numerical methods

Lecture 1: Machine numbers

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- ① Anomalies in computer arithmetic
- ② Machine numbers: a model of floating point numbers

- ➊ Anomalies in computer arithmetic
- ➋ Machine numbers: a model of floating point numbers

What is the value of $\sin(\pi)$?

1.224646799147353e-016

What is the value of $\sum_{k=1}^{+\infty} \frac{1}{k}$?

What is the n th partial sum for some large n ? $\left(\sum_{k=1}^n \frac{1}{k}\right)$

One may sum in one direction or the other ...

For $n = 100'000'000$ we have

18.997896413852555

18.997896413853447

What is the value of $\sqrt{2017} - \sqrt{2016}$?

May be calculated as:

$$\begin{aligned}\sqrt{2017} - \sqrt{2016} &= (\sqrt{2017} - \sqrt{2016}) \cdot \frac{\sqrt{2017} + \sqrt{2016}}{\sqrt{2017} + \sqrt{2016}} = \\ &= \frac{2017 - 2016}{\sqrt{2017} + \sqrt{2016}} = \frac{1}{\sqrt{2017} + \sqrt{2016}}.\end{aligned}$$

Let us try both ways!

0.011134504483941

0.016926965158418

$$a = 10^{-20}, \quad b = 1.$$

What is the value of $a + b$?

1

Does associativity hold?

$$(a + b) - b, \quad a + (b - b) = ?$$

Let us try!

0

1.0000000000000000e-020

What is the value of $\cosh(20) - \sinh(20)$ and $\exp(-20)$?

$$\begin{aligned}\cosh(20) - \sinh(20) &= \frac{\exp(20) + \exp(-20)}{2} - \frac{\exp(20) - \exp(-20)}{2} = \\ &= \exp(-20)\end{aligned}$$

Let us try both ways!

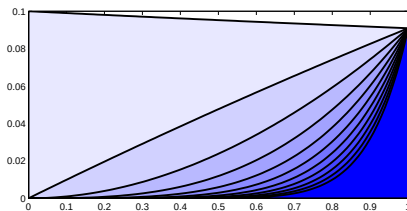
$$\begin{aligned}&0 \\ &2.061153622438558\text{e-}009\end{aligned}$$

What is the value of the following integral?

$$T_n := \int_0^1 f_n(x) = \int_0^1 \frac{x^n}{x+10} dx$$

Hard to find analytically.

We see that it is always positive and tends to zero.



$$\begin{aligned}
 T_n &:= \int_0^1 \frac{x^n}{x+10} dx = \int_0^1 \frac{(x+10-10)x^{n-1}}{x+10} dx = \\
 &= \int_0^1 x^{n-1} dx - 10 \cdot \int_0^1 \frac{x^{n-1}}{x+10} dx = \frac{1}{n} - 10 \cdot T_{n-1}
 \end{aligned}$$

$$T_0 = \int_0^1 \frac{1}{x+10} dx = [\ln(x+10)]_0^1 = \ln(11) - \ln(10) = \ln(1.1)$$

So we have a recursion:

$$T_0 := \ln(1.1), \quad T_n := \frac{1}{n} - 10 \cdot T_{n-1} \quad (n = 1, 2, \dots).$$

Let us calculate T_{20} this way!

Reorder the recursion

$$10 T_{n-1} = \frac{1}{n} - T_n \Leftrightarrow$$

$$T_{n-1} = \frac{1}{10} \cdot \left(\frac{1}{n} - T_n \right)$$

And start from $M \gg n$,

$$T_M := 0, \quad T_{n-1} = \frac{1}{10} \cdot \left(\frac{1}{n} - T_n \right) \quad (n = M, \dots, m+1).$$

Now let us calculate T_{20} this way!

Which method is *stable*?

7.483468021084803e+003
0.004347035818028

Definition:

A *numerical algorithm* is a finite sequence of arithmetic and logical operations.

Definition:

A numerical algorithm is *stable*, if there exists a constant $C > 0$, such that for two input data I_1, I_2 for the output data O_1, O_2 we have

$$\|O_1 - O_2\| \leq C \cdot \|I_1 - I_2\|.$$

Example

The recursion of the Fibonacci(-like) sequence is unstable.

- ① Anomalies in computer arithmetic
- ② Machine numbers: a model of floating point numbers

- In practical and scientific calculations we often need to handle real numbers.
- On the computers we may choose from the elements of a finite set.
- The orders of magnitude may be significantly different.

A model of floating point numbers

Floating point numbers, normalized form: $324 \rightsquigarrow +0.324 \cdot 10^3$.

In binary form: $101000100 \rightsquigarrow +0.101000100 \cdot 2^9$.

In general: $\pm 0. \underbrace{1 \text{ --- } \dots \text{ ---}}_{t \text{ digits}} \cdot 2^k \quad (k^- \leq k \leq k^+).$

Definition: Normalized floating point number

Let $m = \sum_{i=1}^t m_i \cdot 2^{-i}$ with $t \in \mathbb{N}$, $m_1 = 1$, $m_i \in \{0, 1\}$.

The number of the form $a = \pm m \cdot 2^k$ ($k \in \mathbb{Z}$) is called a *normalized floating point number*.

m : mantissa of length' t

k : exponent, $k^- \leq k \leq k^+$

A model of floating point numbers

Notation: $a = \pm[m_1 \dots m_t | k] = \pm 0.m_1 \dots m_t \cdot 2^k$.

Notation: $M = M(t, k^-, k^+)$ is the set of machine numbers for fixed $k^-, k^+ \in \mathbb{Z}$ and $t \in \mathbb{N}$ values. (Generally $k^- < 0$ and $k^+ > 0$.)

Definition: Machine numbers

$$M(t, k^-, k^+) = \left\{ a = \pm 2^k \cdot \sum_{i=1}^t m_i \cdot 2^{-i} : \begin{array}{l} k^- \leq k \leq k^+, \\ m_i \in \{0, 1\}, m_1 = 1 \end{array} \right\} \cup \{0\}$$

In practice we also add: $+\infty, -\infty, \text{NaN}, \dots$

An example of machine numbers

Example

$$M(3, -1, 2) \sim \pm 0.1_ _ \cdot 2^k, \quad (-1 \leq k \leq 2)$$

Elements in case of $k = 0$: 0.100, 0.101, 0.110, 0.111, i.e. $\frac{1}{2}, \frac{5}{8}, \frac{6}{8}, \frac{7}{8}$.

The half of the above in case of $k = -1$, the double in case of $k = 1$, and four times for $k = 2$. (And with negative sign...)

$$\varepsilon_0 = [100|-1] = 0.100 \cdot 2^{-1} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 0.25$$

$$\varepsilon_1 = [101|1] - 1 = 0.101 \cdot 2^1 - 1 = \frac{1}{8} \cdot 2 = \frac{1}{4} = 0.25$$

$$M_\infty = [111|2] = 0.111 \cdot 2^2 = \frac{7}{8} \cdot 4 = \frac{7}{2} = 3.5$$

$$|M| = 2 \cdot 2^2 \cdot 4 + 1 = 33$$

- ① $\frac{1}{2} \leq m < 1$
- ② M is symmetric with respect to 0
- ③ smallest positive value in M :

$$\varepsilon_0 = [100 \dots 0 | k^-] = \frac{1}{2} \cdot 2^{k^-} = 2^{k^- - 1}$$

- ④ the difference of 1 and the next machine number in M :

$$\varepsilon_1 = [100 \dots 01 | 1] - [100 \dots 00 | 1] = 2^{-t} \cdot 2^1 = 2^{1-t}$$

- ⑤ the biggest value in M :

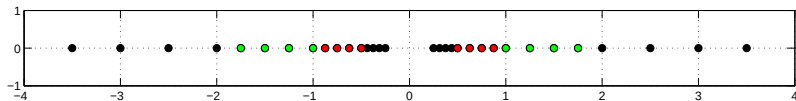
$$\begin{aligned} M_\infty &= [111 \dots 11 | k^+] = 1.00 \dots 00 \cdot 2^{k^+} - 0.00 \dots 01 \cdot 2^{k^+} = \\ &= (1 - 2^{-t}) \cdot 2^{k^+} \end{aligned}$$

- ⑥ the number of elements in M (cardinality):

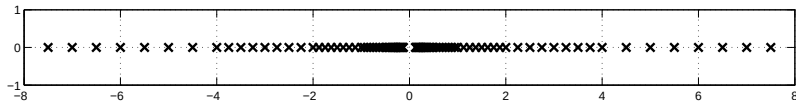
$$|M| = 2 \cdot 2^{t-1} \cdot (k^+ - k^- + 1) + 1$$

Examples of machine numbers

$$M(3, -1, 2)$$



$$M(4, -2, 3)$$



float $\sim M(23, -128, 127)$, double $\sim M(52, -1024, 1023)$

bits, special numbers?

How to convert a number in $\mathbb{R}$ to a set of machine numbers?

Denote by $\mathbb{R}_M$ the set of 'convertible' numbers, i.e.

$$\mathbb{R}_M := \{x \in \mathbb{R} : |x| \leq M_\infty\}.$$

Definition: Input function

The $fl: \mathbb{R}_M \rightarrow M$ function is called *input function* (or floating point conversion function), if

$$fl(x) = \begin{cases} 0 & \text{if } |x| < \varepsilon_0, \\ \tilde{x} & \text{if } \varepsilon_0 \leq |x| \leq M_\infty, \end{cases}$$

where $\tilde{x}$ denotes the machine number closest to x (according to the rules of rounding).

Case of $|x| > M_\infty$? ...

So it already gives an error if we convert real numbers to machine numbers on our computers. . . how much error?

Theorem: Input error

For all $x \in \mathbb{R}_M$:

$$|x - fl(x)| \leq \begin{cases} \varepsilon_0 & \text{if } |x| < \varepsilon_0, \\ \frac{1}{2}|x| \cdot \varepsilon_1 & \text{if } \varepsilon_0 \leq |x| \leq M_\infty. \end{cases}$$

Corollary: Input error

If $\varepsilon_0 \leq |x| \leq M_\infty$, then $\frac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \cdot \varepsilon_1 = 2^{-t}$.

So essentially the error depends on ε_1 , i.e. the mantissa length t .

Conversion to machine numbers

(Remark. What's the error if $|x| > M_\infty$?)

Proof:

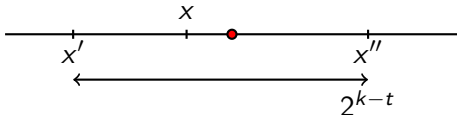
- 1 If $|x| < \varepsilon_0$, then $fl(x) = 0$, so $|x - fl(x)| = |x| < \varepsilon_0$.
- 2 If $|x| \geq \varepsilon_0$ and $x \in M$, then $fl(x) = x$, so $|x - fl(x)| = 0$.
- 3 It is to consider when $|x| \geq \varepsilon_0$ and $x \notin M$.

Enough to consider positive x because of the symmetry.

Find the neighboring machine numbers: $x' < x < x''$ with $x', x'' \in M$.

Let $x = [1_ \dots _ |k]$. What is the distance of x' and x'' ?

If we add 1 in the last digit of x' , then we get x'' . So $x'' - x' = 2^{-t} \cdot 2^k = 2^{k-t}$.



If x is in the first half of the interval, then $fl(x) = x'$, if it is in the second half, then $fl(x) = x''$. Therefore the difference of x and $fl(x)$ is at most half the length of the interval: $\frac{1}{2} \cdot 2^k \cdot 2^{-t}$. I.e.

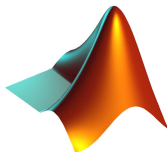
$$|x - fl(x)| \leq \frac{1}{2} \cdot 2^k \cdot 2^{-t}.$$

But for the absolute value of x by its above form

$0.1 \cdot 2^k = \frac{1}{2} \cdot 2^k \leq |x|$ holds, thus the estimation may be continued as:

$$|x - fl(x)| \leq |x| \cdot 2^{-t} = \frac{1}{2} \cdot |x| \cdot \underbrace{2^{1-t}}_{\varepsilon_1} = \frac{1}{2} \cdot |x| \cdot \varepsilon_1.$$





- ① The mentioned anomalies. . .